

Project Title

Marketing Analytics: Customer Engagement & Conversion Analysis

Project Overview

This project analyzes marketing and customer data for **ShopEzip**, an online retail business. The business was experiencing declining conversion rates, reduced customer engagement, and a need to better understand customer feedback.

The objective was to analyze customer journey, marketing engagement, product, and customer review data to identify important trends and provide data-driven recommendations to improve conversion, engagement, and customer satisfaction.

Business Problem

- Declining customer engagement
- Declining conversion rates
- High marketing expenses
- Need for customer feedback analysis

Project Objectives

- Improve conversion rate
- Enhance customer engagement
- Improve customer feedback understanding

Tools & Technologies

SQL | Python | Power BI | Excel

- **SQL:** Data cleaning, transformation, joins, duplicate detection, and analysis
- **Python:** Customer review sentiment analysis using VADER
- **Power BI:** KPI analysis, data modeling, and interactive visualization
- **Excel:** Initial data handling and exploration

Data Used

The analysis used multiple datasets covering:

- Customers
- Products
- Geography
- Customer Journey

- Marketing Engagement
- Customer Reviews

This allowed the analysis to connect customer behavior, product information, marketing engagement, and customer feedback.

Data Preparation

SQL was used to prepare the data for analysis.

Key activities included:

- Combining related datasets using joins
- Identifying duplicate customer journey records
- Handling missing duration values
- Cleaning inconsistent text
- Creating product price categories
- Transforming engagement data into an analysis-ready format

KPI Analysis

Conversion Rate

Measures the percentage of visitors who complete the desired action, such as a purchase.

Overall Conversion Rate: 10%

- Highest: **January — 18.5%**
- Lowest: **May — 4.3%**

CTR — Click-Through Rate

Measures the percentage of views that resulted in clicks.

CTR: 15.37%

Customer Engagement

Analyzed customer interactions using:

- Views
- Clicks
- Likes
- Comments

Views peaked during February and July and declined from August onward.

Customer Feedback

Average customer rating: 3.69/5

- 4-star reviews: **140**
- 5-star reviews: **135**
- Positive sentiment: **275**
- Negative sentiment: **82**

The majority of feedback was positive, but the average rating remained below the project's **4.0 target**.

Dashboard

An interactive Power BI dashboard was developed to analyze marketing conversion, customer engagement, and customer feedback. The dashboard provides filters for year, month, product, and sentiment category, allowing performance to be analyzed from different perspectives.

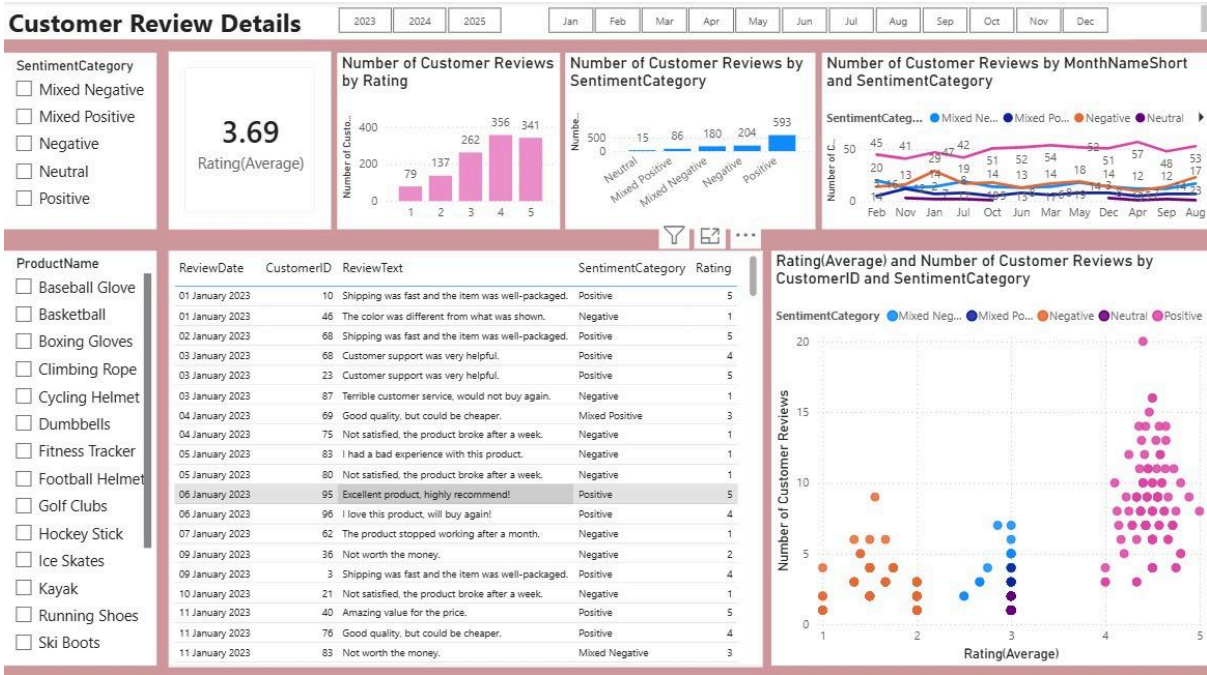
Dashboard 1 – Overview



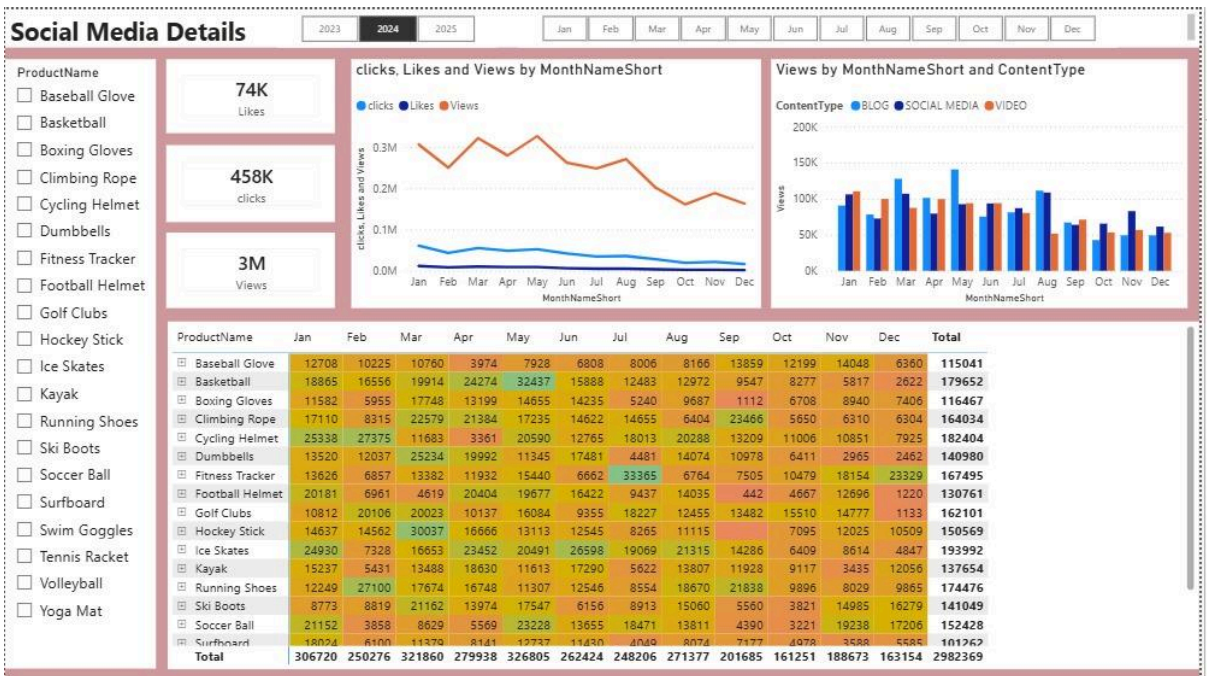
Shows overall conversion rate, social-media engagement, customer ratings, monthly trends, and product-level performance.

Dashboard 2 – Customer Review Details

Analyzes customer ratings and sentiment categories, while also providing detailed customer review-level information.



Dashboard 3 – Social Media Details



Provides detailed analysis of views, clicks, likes, and content-type performance across months and products.

Sentiment Analysis

Customer review text was analyzed using **Python and VADER sentiment analysis**.

The analysis classified reviews into different sentiment categories, helping identify positive, negative, mixed, and neutral customer experiences.

This provided additional context beyond numerical ratings by helping understand **what customers were actually saying**.

Key Findings

- Conversion rates varied considerably across the year.
- January recorded the highest conversion rate at **18.5%**.
- May recorded the lowest conversion rate at **4.3%**.
- Marketing views declined during the later part of the year.
- Blog content generated the highest number of views.
- Customer feedback was generally positive, but the **3.69 average rating** indicated room for improvement.

Recommendations / Solutions

1. Improve Conversion Rates

Focus marketing efforts on **high-performing product categories** such as Kayaks, Ski Boots, and Baseball Gloves. Use seasonal promotions and targeted campaigns during stronger-performing periods to improve conversion during weaker months.

2. Improve Customer Engagement

Refresh the content strategy by experimenting with more engaging formats such as **interactive videos and user-generated content**. Strengthen and optimize calls-to-action, particularly during months where engagement is historically lower.

3. Improve Customer Feedback

Create a feedback loop for **mixed and negative reviews**. Identify recurring customer concerns, address the underlying issues, and use the findings to improve products and customer experience.

Conclusion

The project demonstrates how **SQL, Python, and Power BI can be combined to move from raw business data to actionable insights**. By analyzing conversion, engagement, and customer feedback together, the business can identify performance gaps and make more informed marketing and customer-experience decisions.